

Sancho : The Cursed Conquistador In *Jungle Cruise*

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Figure 1: Partially-formed Sancho tasting the honey with honeybees swarming around

ABSTRACT

In *Jungle Cruise*, Sancho is one of the cursed conquistadors whose partially-decomposed body is composed of honeycombs that the honeybees ride on and has honey dripping off all the time. His topology will change throughout the sequence and need to interact with surrounding environment, which creates unique visual and technical challenges. To achieve this we established a procedural workflow from modeling to rendering that is both efficient and art-directable.

KEYWORDS

Dynamics/Simulation, Procedural Geometry, Production

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1 TURN INTO CURSED SANCHO

In the story, Sancho can split into swarm of bees and forming back to normal human. Which means we need to covert him into fully procedural geometries. And there are honeybees moving around on the surface, and then flocking around.

2 PROCEDURAL HONEYCOMBS

Since Sancho is composed of honeycombs, our first step is converting the model into multiple layers of packed hexagons. Utilizing triangulation with fixed edge length and the property of dual shape. We can easily convert any surface into hexagons. Then extrude inward and repeat the process until the volume is filled. Point clouds with attributes were created from these packed hexagons pivot points at neutral position, which provides efficient control and fast visualization.

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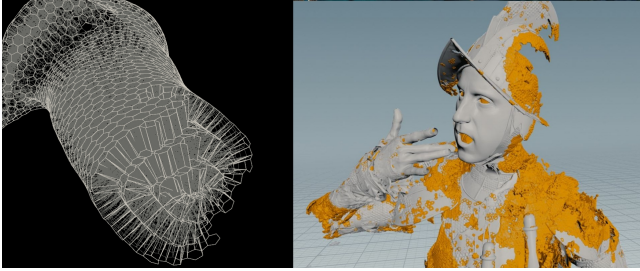


Figure 2: Turn arm into hexagons (left) and partially formed Sancho (right).

3 TRANSITION FROM ASSET TO CURSED

With “rebuild” attribute on the point clouds, which defines the stages during the transition from normal to cursed Sancho. These stages starting from null, base hexagons, extrude honeycomb volume, filled with honey, closed honeycomb lid, finally deform to normal Sancho. All the stages mapping to specific procedural shapes on single honeycomb, which can be turn into signed distance fields and interpolated seamlessly with rebuild attribute. Wave propagation algorithm was used on the point clouds to drive the rebuild attribute to generate the forming effect for Sancho. Cavities with partially formed honeycombs really make Sancho being badly cursed.



Figure 3: Merged result of honey layers

4 HONEY SIMULATION

Multiple layers of honey are meshed together to form the final yummy look. Most of the layers are procedural, emit from honeycombs with open lids then advect along surface field to generate streaming honey on the body. Dripping honey was achieved by custom solver which generates breakable sticky particle chains. The length, width and break timing are fully art-directable, which is the major difference from standard viscous flip simulation. We also have extremely fast moving shots, the solver could produce long breakable honey strings matching the reference very well. Finally, a radioactive shape will be generated from each string to form the organic look. Few particles from these steps are selected and will be rendered as bubbles inside the honey.

5 HONEYBEES SIMULATION

Honeybees simulations are made of 3 layers, bees crawling around the body, bees stacking on body and bees following Sancho. For



Figure 4: Crawling, stacking and flocking bees

crawling layer, areas of emission are selected around honeycomb then particles moving along surface guided by curl field are simulated at origin. Then deformed to follow Sancho’s animated mesh. Additional extruded geometries are added on areas where bees stack up. With soft body dynamics to generate jiggling effect. Bees are added in layers to fill up the geometry for stacking. Based on the emission frame and position of each bee, we have another set of points that travel from around Sancho to his body and land at the same spot. Which provides smooth transition between flying and crawling bees.



Figure 5: Transition between different Materials

6 LIGHTING AND RENDERING

From the honeycombs to Sancho, we defined several materials mapping to each stage during the transition, which can be blended smoothly and driven by the same “rebuild” attributes. Few more attributes on the geometries also help to create natural randomness for colour and surface variations for each honeycomb. It was very important that the shaders were using physical attributes like transmission absorption for a believable light transport through the different layers of honey. With those attributes in place we could light Sancho in a natural way and still make him look believable and realistic. Especially when Sancho was backlit, all the layers of honeycomb made it really interesting to look at.

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